Real Numbers and Rational Exponents

Concept

Real numbers include rational and irrational numbers. Fractional powers represent roots.

Classification

Rational: integer fractions; irrational: non-repeating, non-terminating decimals.

Fractional power

$a^{m/n} = (\sqrt[n]{a})^m$ for $a > 0$; reduce m/n first.

Negative power

$a^{-m/n} = 1/a^{m/n}$, on its non-zero real domain.

Root sign

$\sqrt{(x^2)} = |x|$; odd roots can accept negative inputs.

Example 1

Evaluate $32^{3/5}$ and $81^{-1/2}$.

Solution

1. $32^{3/5} = (\sqrt[5]{32})^3 = 2^3 = 8$.

2. $81^{-1/2} = 1/\sqrt{81} = 1/9$.

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Example 2

Simplify $\sqrt{(49x^2)}$, then evaluate it at $x = -2$.

Solution

1. $\sqrt{(49x^2)} = 7|x|$, because the square root is non-negative.

2. At $x = -2$: $7|-2| = 14$.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Evaluate $64^{2/3}$.

2. Evaluate $25^{-3/2}$.

3. Simplify $\sqrt{(49x^2)}$.

4. For $x > 0$, simplify $x^{5/6}/x^{1/3}$.

5. Is $\sqrt[3]{(-8)}$ real? Is $\sqrt{(-8)}$ real?

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Practice answers | check after trying

1. $(\sqrt[3]{64})^2 = 4^2 = 16$.

2. $1/(\sqrt{25})^3 = 1/125$.

3. $7|x|$.

4. $x^{5/6 - 2/6} = x^{1/2}$.

5. $\sqrt[3]{(-8)} = -2$ is real; $\sqrt{(-8)}$ has no real value.